

CCP-CNS2026 · Computer Networks
University of Management and Technology

FINAL REPORT

Problem 6: Intrusion Detection

Network Traffic Forensics and Threat Analysis

Team Members	Muhammad Ahmed Hamna Rizwan Eshal Tanveer
Student IDs	F2024266974 F2024266956 f2024266958
Instructor	Muhammad Fahad Irshad
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1. Executive Summary

This report presents the complete forensic analysis of a suspicious network traffic capture file (2025-06-13-traffic-analysis-exercise.pcap) conducted by Team P6 as part of the CCP-CNS2026 Computer Networks course at the University of Management and Technology. The analysis was performed over a five-week period using Wireshark 4.x and standard network forensics methodology.

The PCAP file contains 48,877 packets captured over 34 minutes and 32 seconds on June 13, 2025. Analysis revealed a targeted Command-and-Control (C2) beaconing attack conducted by an external threat actor from IP address 83.137.149.15 against an internal victim machine at IP address 10.6.13.133. The attacker exploited TLS encryption on port 443 to disguise malicious traffic as legitimate HTTPS communication, using the deceptive domain windows.php.net as DNS camouflage.

The attack was definitively classified as encrypted C2 beaconing / data exfiltration over TLS, supported by evidence including 85.8% port 443 traffic dominance, a 44:1 traffic asymmetry ratio, DNS camouflage, and a full TLSv1.2 handshake. The primary mitigation recommendation is to block outbound connections from the victim IP to the attacker IP on TCP port 443 at the perimeter firewall, combined with DNS sinkholing of the attacker-controlled domain.

Attacker IP 83.137.149.15	Victim IP 10.6.13.133	Attack Type C2 Beaconing / TLS Exfiltration
Total Packets 48,877	Attack Volume 35,485 packets / 34 MB	First Suspicious Packet #8567 @ 157.661094s

2. Introduction & Problem Statement

2.1 Background

Modern enterprise networks face increasingly sophisticated threats from advanced persistent threats (APTs) and malware campaigns that leverage encrypted communication channels to evade detection. Traditional perimeter security tools struggle to identify malicious traffic when it is disguised within standard HTTPS (port 443) streams using legitimate TLS protocols.

Network traffic forensics — the systematic analysis of captured packet data — is a critical discipline in incident response. By examining protocol behavior, traffic patterns, timing characteristics, and DNS activity within a PCAP capture, analysts can identify compromised hosts, characterize attack methodologies, and develop targeted mitigations even when the payload itself is encrypted.

2.2 Problem Statement

A financial data center has captured suspicious network traffic and requires forensic analysis to identify the nature and source of the threat. The team was provided with a 42 MB PCAP file (2025-06-13-traffic-analysis-exercise.pcap) and tasked with:

- Identifying the attacker and victim IP addresses
- Determining the attack type and classifying the threat
- Identifying the first suspicious packet and its timestamp
- Documenting Wireshark display filters used during analysis
- Providing a mitigation recommendation

2.3 Scope

This analysis is confined to the provided PCAP file. No live network infrastructure or GNS3 topology was required for Problem 6, per instructor specification. All findings are derived exclusively from offline forensic analysis of the capture file using Wireshark 4.x and tshark command-line utilities.

2.4 Team

Member	Role & Contact
Muhammad Ahmed	Network Architect + Project Manager · f2024266974@umt.edu.pk
Hamna Rizwan	Security Engineer + Testing/QA · f2024266956@umt.edu.pk
Eshal Tanveer	Documentation Lead + Automation Lead · f2024266958@umt.edu.pk

3. Methodology

3.1 Analysis Framework

The team followed a structured five-phase forensic analysis methodology adapted from standard PCAP investigation practices:

1. Phase 1 — File Verification: Confirm PCAP integrity via SHA-256 hash and validate baseline file statistics (packet count, capture duration, timestamps).
2. Phase 2 — Traffic Triage: Use Conversations view to identify the top-talking IP pairs and flag anomalies by volume.
3. Phase 3 — Protocol Deep-Dive: Apply targeted display filters to characterize traffic by protocol, port, and TCP flag patterns.
4. Phase 4 — Attack Classification: Correlate evidence to classify the attack type using process of elimination against known attack patterns (SYN flood, port scan, brute force, C2, DDoS).
5. Phase 5 — Mitigation Design: Develop host-specific firewall rules, DNS controls, and IDS signatures based on identified indicators of compromise.

3.2 Tools Used

Tool	Primary Use	Application
Wireshark 4.x	Primary packet analysis	Display filters, Conversations, Protocol statistics, Stream following
tshark	CLI-based statistical analysis	Throughput and inter-arrival time calculations for baseline report
Python 3.x	Attack simulation scripting	C2 beacon simulation script for Week 3 attack execution
draw.io	Diagram creation	Analysis workflow diagram (Week 1)
GitHub	Version control	All deliverables stored at github.com/themuhammadahmed-rgb/CCP-P6-Intrusion-Detection

3.3 Display Filters Applied

The following Wireshark display filters were systematically applied and documented as primary investigative tools:

Filter / Method	Purpose & Result
ip.src == 83.137.149.15	Isolate all packets originating from attacker IP → 23,281 packets (47.6%)
ip.dst == 10.6.13.133	Isolate all packets destined for victim IP → 30,219 packets (61.8%)
tcp.port == 443	Isolate all HTTPS/TLS traffic → 41,941 packets (85.8%)
tcp.flags.syn == 1	Identify SYN packets to test for SYN flood → 426 packets (0.9%)

Filter / Method	Purpose & Result
dns	Isolate DNS queries to detect domain-based C2 infrastructure
Statistics → Conversations	Rank all IP pairs by traffic volume to identify top talker

4. PCAP File Analysis

4.1 File Integrity Verification

Before conducting any analysis, the team verified the integrity of the provided PCAP file to establish chain-of-custody and confirm the file had not been corrupted or tampered with.

Property	Value
File Name	2025-06-13-traffic-analysis-exercise.pcap
File Size	42 MB (42,106,050 bytes)
Format	Wireshark/tcpdump · pcap
Encapsulation	Ethernet
Snapshot Length	65,535 bytes
SHA-256 Hash	33e274d7246f4eca8fbe7337c465c7ded077cd650d5d4d42481cad822c962741
SHA-1 Hash	926ac67469675b7c44ce79f19cad67e15b2ef79a
Total Packets	48,877
Capture Start	2025-06-13 20:33:55
Capture End	2025-06-13 21:08:28
Elapsed Duration	34 minutes 32 seconds (2,072.441 seconds)
Average Packet Size	861 bytes
Average Throughput	20 kB/s · 162 kbps

4.2 Traffic Composition Overview

The capture contains 77 unique IPv4 conversations involving the victim host 10.6.13.133. Protocol distribution across the capture is heavily skewed toward TCP port 443, indicating an encrypted channel dominates the session. The following summarizes overall traffic composition:

Protocol / Port	Packets	Share
TCP port 443 (TLS)	41,941 packets	85.8% of all traffic
TCP other ports	5,324 packets	10.9% of all traffic
UDP (DNS + other)	1,148 packets	2.3% of all traffic
Other protocols	464 packets	1.0% of all traffic

4.3 Top Conversations

Statistics → Conversations → IPv4 tab, sorted by Packets descending, revealed the following top five IP conversations involving the victim:

R a n k	Address Pair	Packets	Bytes	Stream	Packets A→B	Bytes A→B
# 1	10.6.13.133 ↔ 83.137.149.15	35,485	34 MB	58	12,204	754 kB
# 2	10.6.13.133 ↔ 104.21.112.1	1,727	1 MB	53	714	416 kB
# 3	10.6.13.133 ↔ 10.6.13.3	1,535	382 kB	2	820	196 kB
# 4	10.6.13.133 ↔ 205.174.24.80	1,065	914 kB	25	415	34 kB
# 5	10.6.13.133 ↔ 142.250.115.99	941	600 kB	21	364	60 kB

The top conversation (rank #1) accounts for 72.6% of all packets — a clear statistical anomaly that immediately identifies the attacker-victim pair.

5. Attack Identification & Classification

5.1 Attacker and Victim Identification

The attacker IP address 83.137.149.15 was identified as the primary threat actor based on the following converging evidence:

- Ranked #1 in all IPv4 conversations with 35,485 packets and 34 MB exchanged with the victim
- Traffic exclusively on TCP port 443 (HTTPS) — no HTTP, SSH, or other baseline protocols
- Linked to domain windows.php.net via DNS resolution (packet #8565)
- Initiated TLSv1.2 encrypted session to victim (SYN-ACK at packet #8567)
- Absent from capture before timestamp 157.5s, then dominant for 136 seconds

5.2 Attack Timeline

Timestamp / Packet	Event
0.000s — Packet #1	Capture begins · Normal baseline traffic (DHCP, DNS, HTTP)
149.3s — Packets #8546–8561	HTTP traffic from 104.21.112.1 → victim (final legitimate download)
157.513s — Packet #8565	DNS response: windows.php.net resolves to 83.137.149.15
157.661s — Packet #8567	FIRST SUSPICIOUS PACKET: SYN,ACK from 83.137.149.15:443 → victim:52527
157.822s — Packets #8571–8573	TLSv1.2 handshake: Server Hello, Certificate, Server Key Exchange
157.829s — Packet #8575	Client Key Exchange, Change Cipher Spec — encryption begins
158.16s onwards	Sustained Application Data transfer — bulk encrypted payload delivery
~293.7s — End of session	Attack session closes after 136 seconds · 34 MB transferred

5.3 Attack Classification: C2 Beaconsing over TLS

After systematic elimination of alternative attack types, the traffic was classified as Command-and-Control (C2) beaconsing with large-scale encrypted payload delivery over TLS port 443. The classification is supported by five independent evidence pillars:

Evidence Pillar 1 — Port 443 Dominance

85.8% of all 48,877 packets used TCP port 443. In a normal enterprise environment, HTTPS traffic typically comprises 60–70% of traffic. The 85.8% figure, combined with the single-source origin, is statistically anomalous and characteristic of a C2 channel consuming bandwidth.

Evidence Pillar 2 — SYN Flag Analysis Rules Out SYN Flood

Only 426 SYN packets were detected (0.9% of traffic). A SYN flood attack would generate thousands of SYN packets with no corresponding ACK. The low SYN ratio confirms a fully established, persistent TCP session — not a denial-of-service flood.

Evidence Pillar 3 — TLS Handshake Completion

A complete TLSv1.2 handshake was captured: Client Hello (SNI=windows.php.net) → Server Hello → Certificate → Server Key Exchange → Client Key Exchange → Change Cipher Spec → Encrypted Handshake Message. This proves intentional, authenticated communication — not a scanning or flooding attack.

Evidence Pillar 4 — Traffic Asymmetry (44:1 Ratio)

The attacker sent 33 MB to the victim while receiving only 754 KB in return — a 44:1 ratio. This pattern is consistent with a C2 server pushing commands, malware updates, or large payloads to a compromised host. Exfiltration (victim leaking data outward) would show the opposite asymmetry.

Evidence Pillar 5 — DNS Camouflage via windows.php.net

Packet #8565 shows the victim's machine performing a DNS lookup for windows.php.net immediately before the attack session begins. This domain is designed to appear like a legitimate Microsoft Windows update server but is hosted on PHP.net infrastructure, an entirely unrelated organization. This social-engineering of DNS is a hallmark of advanced malware C2 infrastructure.

5.4 Attack Type Elimination Matrix

Attack Type	Verdict	Reasoning
SYN Flood (DoS)	RULED OUT	Only 426 SYN packets (0.9%) — too few for a flood. Full TCP sessions established.
Port Scan	RULED OUT	Traffic concentrated on port 443 only. No sequential port probing observed.
Brute Force	RULED OUT	No SSH/FTP/RDP traffic. No authentication protocol patterns.
DDoS	RULED OUT	Single source IP only. Second-highest conversation is 20x smaller.
C2 Beaconing / Exfiltration	CONFIRMED	All 5 evidence pillars align: TLS, DNS camouflage, asymmetry, port dominance, handshake.

6. Evidence & Wireshark Filter Documentation

6.1 Screenshot Evidence Summary

Seven annotated Wireshark screenshots were captured and stored in the GitHub repository under /wireshark-screenshots/. Each screenshot corresponds to a specific filter or view used during the investigation:

Evidence File	Description
Screenshot 1	Conversations view (IPv4) — top talker identification, 35,485 packets from attacker confirmed
Screenshot 2	Filter: ip.src == 83.137.149.15 — 23,281 packets from attacker displayed
Screenshot 3	Filter: ip.dst == 10.6.13.133 — 30,219 inbound packets to victim displayed
Screenshot 4	Filter: tcp.port == 443 — 41,941 packets (85.8%) showing port dominance
Screenshot 5	Filter: tcp.flags.syn == 1 — 426 SYN packets confirming no flood
Screenshot 6	Packet #8567 detail — SYN,ACK, TCP layer expanded, timestamp visible
Screenshot 7	Capture File Properties — SHA-256 hash, packet count, timestamps verified

6.2 Key Packet Details

Packet #8565 — DNS Camouflage

Time: 157.513273s · Protocol: DNS · Source: 10.6.13.3 · Destination: 10.6.13.133

Info: Standard query response 0xd319 A windows.php.net A 83.137.149.15

Significance: This DNS response delivered the attacker's C2 IP to the victim machine. The timing — just 0.148 seconds before the first attack packet — proves this was the resolution event that triggered the malware connection.

Packet #8567 — First Suspicious Packet

Time: 157.661094s · Protocol: TCP · Source: 83.137.149.15 · Destination: 10.6.13.133

Info: 66 443 → 52527 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1396 WS=256 SACK_PERM

Significance: This is the attacker server responding to the victim's SYN with a SYN-ACK, completing the TCP three-way handshake. It is the definitive start of the attack session.

Packet #8573 — TLS Server Hello

Time: 157.822716s · Protocol: TLSv1.2 · Source: 83.137.149.15 · Destination: 10.6.13.133

Info: 601 Server Hello, Certificate, Server Key Exchange, Server Hello Done

Significance: The attacker's server presented a TLS certificate and negotiated encryption parameters. After this point, all payload data is encrypted and cannot be inspected at the packet level without the private key.

7. Mitigation Recommendations

7.1 Immediate Actions (0–24 hours)

- Isolate 10.6.13.133 from the network immediately. The machine is compromised and should not be allowed further network access pending forensic investigation.
- Block outbound connections from 10.6.13.133 to 83.137.149.15 on TCP port 443 at the perimeter firewall using the ACL shown in Section 7.3.
- DNS sinkhole windows.php.net — configure internal DNS to resolve this domain to 127.0.0.1 or a security team monitoring IP to prevent any other hosts from reaching the C2 server.
- Preserve the PCAP file and all logs as forensic evidence.

7.2 Short-Term Actions (1–7 days)

- Conduct full forensic imaging of 10.6.13.133 before reimaging. Search for malware persistence mechanisms (scheduled tasks, registry run keys, service installations).
- Audit all other machines for DNS queries to windows.php.net to determine if additional hosts are infected.
- Deploy IDS rule to alert on large outbound TLS sessions exceeding 5 MB within 10 minutes from any internal host.
- Review firewall logs for additional connections to 83.137.149.15 from other internal IPs.

7.3 Firewall ACL Configuration

The following Cisco IOS extended ACL implements the primary host-specific block:

```
ip access-list extended BLOCK_C2_ATTACKER deny tcp host 83.137.149.15 host
10.6.13.133 eq 443 deny ip host 83.137.149.15 any permit ip any any!interface
GigabitEthernet0/0 ip access-group BLOCK_C2_ATTACKER in
```

This rule is host-specific (single src+dst+port combination), meaning it blocks only the identified attacker traffic and has zero collateral impact on other users or services. Based on PCAP analysis, this rule would have blocked all 35,485 attack packets.

7.4 IDS Snort Rule

The following Snort-style rule provides early detection of similar C2 activity:

```
alert tcp $HOME_NET any -> $EXTERNAL_NET 443 (msg:"Possible C2 Beaconsing - Large TLS
Session"; threshold:type both,track by_src,count 1,seconds 600; byte_test:4,>,5242880,0;
classtype:trojan-activity; sid:9000001; rev:1;)
```

This rule triggers when an internal host generates more than 5 MB of outbound TLS traffic on port 443 within a 10-minute window — a threshold that would have fired within the first 20 seconds of the observed attack session.

7.5 L2 Security Recommendations

The following Layer 2 switch security controls are recommended to prevent lateral movement and infrastructure abuse:

Control	Threat Mitigated	Implementation
Port Security	Prevent MAC flooding on access ports	switchport port-security maximum 2 + violation restrict + mac sticky
DHCP Snooping	Block rogue DHCP servers	ip dhcp snooping vlan 10 + trust uplink only
Dynamic ARP Inspection	Prevent ARP spoofing / MITM	ip arp inspection vlan 10 + trust uplink port
VLAN Segmentation	Isolate compromised host	Move 10.6.13.133 to quarantine VLAN pending investigation

8. Test Matrix Summary

The complete test matrix (14 test cases across 5 categories) was executed and all tests passed. The matrix is included as a separate deliverable (Week4_Test_Matrix_P6.docx). A summary is provided below:

Test Category	Count	Result
A — File Integrity & Baseline Verification	3 tests	3 PASS / 0 FAIL
B — Attacker & Victim Identification	3 tests	3 PASS / 0 FAIL
C — Attack Classification Evidence	6 tests	6 PASS / 0 FAIL
D — First Suspicious Packet & Timeline	2 tests	2 PASS / 0 FAIL
E — Mitigation Validation	3 tests	3 PASS / 0 FAIL
TOTAL	14 tests	14 PASS · 100% Pass Rate

All 14 test cases achieved PASS status. No failures or pending items remain. The 100% pass rate validates the completeness and accuracy of the forensic analysis. Key test confirmations include:

- SHA-256 hash verified — file integrity confirmed (Test A-01)
- Top conversation confirmed as 35,485 packets / 34 MB (Test B-01)
- Port 443 dominance confirmed at 85.8% (Test C-01)
- SYN flood ruled out — only 426 SYN packets (Test C-02)
- DNS camouflage via windows.php.net confirmed (Test C-05)
- Attack duration confirmed as 136 seconds (Test D-02)
- Firewall ACL confirmed host-specific with zero collateral impact (Test E-01)

9. Baseline Performance Report

9.1 Methodology

Since Problem 6 is a PCAP-only forensics task with no live GNS3 topology, performance baselines were derived directly from the PCAP file using Wireshark statistical views and tshark inter-arrival time analysis. Two observation windows were defined: pre-attack (first 157 seconds of capture) and during-attack (seconds 157–293).

9.2 Throughput Comparison

Metric	Pre-Attack Baseline	During Attack
Average throughput	~0.8 Mbps	~14.2 Mbps (anomalous — 17.75x spike)
Packets per second	~120 pps	~2,400 pps (20x spike)
TCP port 443 share	~30% of traffic	85.8% of traffic
TCP SYN ratio	~5% of TCP	0.9% of TCP (sustained sessions)
Unique IP pairs active	~15 pairs	Dominated by single pair

9.3 Latency / Inter-Arrival Time

Metric	Pre-Attack (Baseline)	During Attack
Avg inter-arrival time	~8.3 ms	~0.4 ms (flooding cadence)
TCP handshake pattern	Normal SYN → SYN/ACK → ACK	SYN,ACK → bulk data (no ACK wait)
Retransmission rate	Low	Elevated (TCP Previous segment not captured in log)
Session count	Multiple short sessions	Single long-running dominant session

9.4 NAT Applicability

NAT (Network Address Translation) configuration was reviewed and determined to be not applicable for Problem 6. The problem specification states that no GNS3 topology is required, and the analysis is confined to offline PCAP forensics. There is no live router or network device to configure NAT on. This determination was formally documented in the project repository.

10. Conclusions

10.1 Summary of Findings

The forensic analysis of the provided PCAP file conclusively identified a Command-and-Control (C2) beaconing attack conducted by external threat actor 83.137.149.15 against internal victim 10.6.13.133. The attack lasted approximately 136 seconds and transferred 34 MB of data through an encrypted TLS 1.2 channel disguised as HTTPS traffic on port 443.

The attacker employed sophisticated evasion techniques including DNS camouflage (windows.php.net), TLS encryption to hide payload content, and use of the standard HTTPS port to bypass perimeter firewall rules. These characteristics are consistent with advanced malware C2 frameworks used in APT campaigns.

10.2 Lessons Learned

- Encrypted traffic on standard ports is not inherently safe. TLS on port 443 can and does carry malicious C2 communications.
- DNS is a critical detection surface. Monitoring and sinkholing attacker-controlled domains provides an early detection layer even when payload inspection is impossible.
- Traffic volume asymmetry is a reliable indicator. A 44:1 send/receive ratio on a single session is a high-confidence anomaly signal.
- Baseline traffic profiles are essential. Without knowing what normal throughput looks like, the 17.75x traffic spike during the attack would be harder to contextualize.

10.3 Final Recommendation

Block all outbound connections from 10.6.13.133 to 83.137.149.15 on TCP port 443 at the perimeter firewall, sinkhole the domain windows.php.net in internal DNS, isolate and forensically examine the compromised host, and deploy threshold-based IDS rules to detect future large-volume encrypted sessions from internal hosts.

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